

Integrated Genomic, Transcriptomic, and Neoantigen Prediction Analysis in Hepatocellular Carcinoma (TCGA-LIHC)

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1. Introduction

Hepatocellular carcinoma (HCC) accounts for approximately 90% of primary liver cancers and remains one of the leading causes of cancer-related mortality worldwide. HCC predominantly arises within the context of chronic liver inflammation, cirrhosis, viral hepatitis infections (HBV and HCV), alcohol abuse, and metabolic dysfunction-associated steatohepatitis (MASH). The genomic landscape of HCC is characterized by high inter- and intra-tumor heterogeneity driven by hallmark mutations in tumor suppressors (e.g. *TP53*), oncogenic developmental signaling pathways (e.g. Wnt/ β -catenin activation via *CTNNB1*), and telomere maintenance mechanisms.

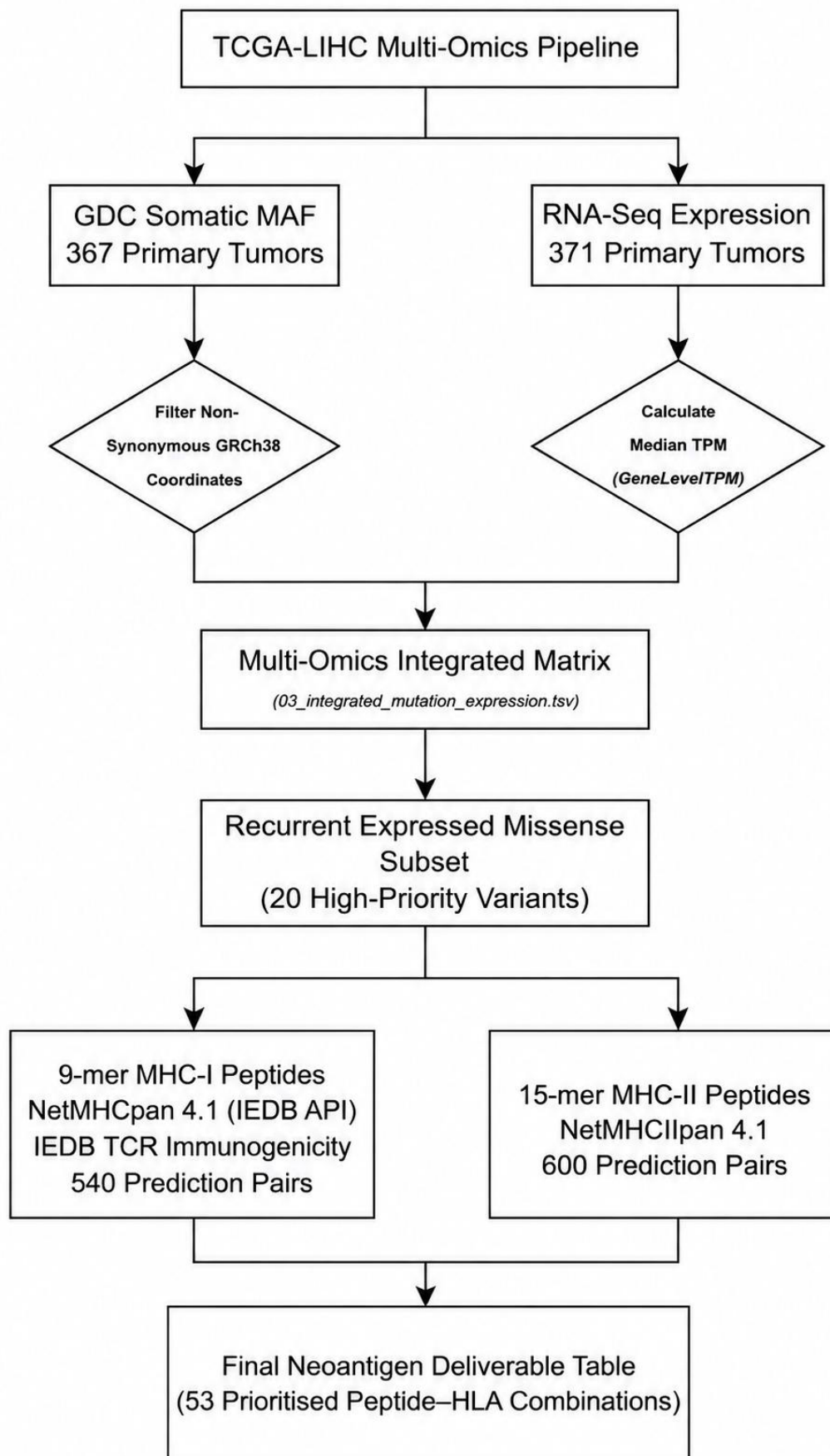
While immune checkpoint inhibitors (ICIs) targeting PD-1/PD-L1 and CTLA-4 have reshaped systemic therapy for unresectable HCC, clinical response rates remain moderate (~15–30%). A critical determinant of anti-tumor T-cell reactivity is the generation of neoantigens—novel immunogenic peptides generated from somatic non-synonymous mutations that are processed, loaded onto Major Histocompatibility Complex (MHC) class I or class II molecules, and presented on the cell surface for T-cell receptor (TCR) recognition.

Predicting true immunogenic neoantigens requires a multi-omics approach: somatic mutations must be transcriptionally expressed within the tumor, processed into peptide fragments, bind stably to patient HLA alleles with higher affinity than their wild-type counterparts, and elicit productive TCR stimulation. In this study, we established an automated, reproducible bioinformatics workflow integrating somatic mutation profiles, RNA-seq expression levels (TPM), and peptide-MHC binding predictions across 367 primary tumor samples from the TCGA-LIHC cohort.

2. Data Sources and Pipeline

All raw genomic and transcriptomic datasets were obtained from the National Cancer Institute Genomic Data Commons (NCI-GDC) and Bioconductor repositories.

Data Modality	Repository / Platform	Primary Identifier / Experiment	Reference Version	Output Dimension
Somatic Mutations	NCI-GDC	TCGA-LIHC MAF (Masked Somatic)	GRCh38 / hg38	367 primary tumors
Transcriptomics	Bioconductor (<i>curatedTCGAData</i>)	<i>LIHC_RNASeq2Gene</i>	RSEM TPM	371 primary tumors
Protein Sequences	UniProtKB / Ensembl	UniProt REST API / VEP	VEP v104 / Release 2026_01	Canonical transcripts
Epitope Prediction	IEDB Tools API	NetMHCpan 4.1 / NetMHCIIpan 4.1	Version 4.1 BA	1,140 peptide – HLA comparison rows



3. Methods

3.1. Somatic Mutation Processing and Binary Matrix Construction

MAF records were filtered using *01_prepare_mutations.R* and *02_build_mutation_matrix.R*. Variant filtering retained protein-coding, high-confidence somatic variants matching TCGA primary solid tumor barcodes (sample code *01*). Non-synonymous classes (missense, nonsense, frame-shift, in-frame indels, and splice-site mutations) were formatted into standard HGVS notation (*c.DNA* and *p.Protein*). A binary mutation-by-sample matrix was constructed where rows represent unique variants ($N = 32,933$) and columns represent primary tumor samples ($N = 367$), with values populated as 1 (present) or 0 (absent).

3.2. Expression Processing and Median Aggregation

Transcriptomic data were obtained directly in R through curatedTCGADData version 2.1.1 using the RNASeq2Gene assay, documented by the source as RSEM TPM. Primary tumour samples were retained, while 50 solid-tissue normal samples and 2 recurrent tumour samples were excluded. Because the mutation and expression cohorts were not fully matched, GeneLevelTPM was calculated as the median expression value across 371 primary tumour samples.

$$\text{GeneLevelTPM} = \text{median}(TPM_{i1}, TPM_{i2}, \dots, TPM_{im})$$

across all primary tumor samples $m = 371$

3.3. Dataset Integration and Quality Control

The binary mutation matrix was joined with the *GeneLevelTPM* summary table in *04_integrate_datasets.R*. Quality-control checks confirmed complete binary mutation calls. Of 32,933 mutation rows, 29,947 (90.93%) were matched to the RNA-seq dataset, while 2,986 rows (9.07%) had no matched expression value and were retained as NA. Overall, 11,275 unique mutated genes had an available GeneLevelTPM value.

3.4. Variant-to-Peptide Mapping & Peptide Windowing

For the advanced neoantigen component (*06_select_missense_mutations.R* through *10_generate_mutant_peptides.R*), 20 recurrent missense mutations (≥ 2 tumor samples, *GeneLevelTPM* > 1) were selected across critical HCC oncogenes and driver genes (*CTNNB1*, *TP53*, *PIK3CA*, *EEF1A1*, *GNAS*). Full-length reference protein sequences were retrieved from UniProt and validated with 100% concordance against reference amino acid identities and protein lengths.

MANE Select transcripts were preferred for protein annotation. When a MANE transcript was unavailable for *GNAS*, a validated protein-coding transcript from the existing GDC/VEP annotation was used.

For each missense variant at position m , sliding windows generated all possible mutation-containing 9-mers (MHC Class I) and 15-mers (MHC Class II), paired with matching wild-type sequences differing strictly by a single amino acid:

- **9-mer generation:** $k = 9$ sliding windows per variant ($9 \times 20 = 180$ pairs $\rightarrow 360$ total peptides).
- **15-mer generation:** $k = 15$ sliding windows per variant ($15 \times 20 = 300$ pairs $\rightarrow 600$ total peptides).

3.5. HLA Binding and Immunogenicity Scoring

Peptide binding was evaluated against a representative fixed panel:

HLA Class	Allele Panel	Prediction Tool / Method	Source
Class I	HLA-A*01:01	NETMHCpan 4.1 BA	IEDB API
	HLA-A*02:01		
	HLA-A*03:01		
Class II	HLA-DRB1*01:01	NETMHCIIpan 4.1 BA	
	HLA-DRB1*04:01		

Binding metrics included half-maximal inhibitory concentration (IC_{50} in nM) and Percentile Rank. Class I binders were categorized as Strong ($IC_{50} \leq 50$ nM), Weak ($50 < IC_{50} \leq 500$ nM), or Non-Binder (> 500 nM). Class II binders were classified by Percentile Rank ($\leq 2\%$ Strong, $\leq 10\%$ Weak).

To quantify preferential presentation of the neo-epitope over self-peptides, differential binding metrics were calculated:

$$\Delta Affinity = IC_{50}^{WildType} - IC_{50}^{Mutant}$$

$$AffinityFoldChange = \frac{IC_{50}^{WildType}}{IC_{50}^{Mutant}}$$

Because lower IC_{50} values reflect stronger affinity, a positive $\Delta Affinity$ and $AffinityFoldChange > 1$ indicate superior mutant binding.

Immunogenicity scores for mutant 9-mer peptides were calculated using the IEDB Next-Generation API version 0.1 Class I Immunogenicity predictor, with positions 1, 2, and 9 masked. Higher scores indicate greater predicted immunogenic potential but do not represent a calibrated probability of T-cell activation.

4. Results

4.1. Mutation Landscape in TCGA-LIHC

Across 367 primary TCGA-LIHC solid tumor samples analyzed, somatic variant filtering identified a mutational landscape dominated by classic hepatocellular carcinoma drivers (**Figure 1**). The tumor suppressor TP53 was the most frequently mutated gene ($N = 98$, 26.70%), followed closely by the Wnt/ β -catenin pathway oncogene *CTNNB1* ($N = 94$,

25.61%) and the giant structural protein TTN ($N = 89$, 24.25%). Additional recurrent lesions were identified in *MUC16* (16.08%), *ALB* (11.99%), and *PCLO* (11.17%), with *RYR2*, *APOB*, *CSMD3*, and *LRP1B* mutated in approximately 8–9% of the cohort.

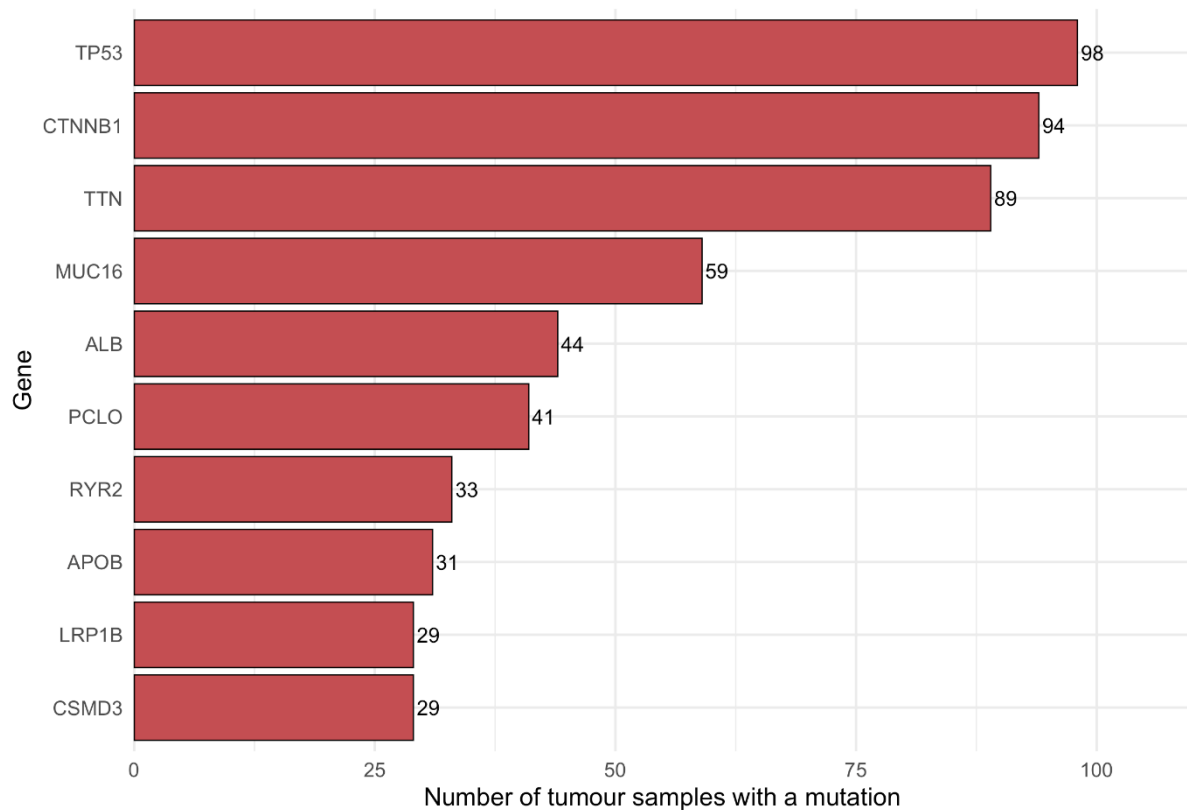


Figure 1. Top 10 most frequently mutated genes in TCGA-LIHC ($N = 367$).

4.2. Gene Expression Profile of Mutated Genes

To evaluate the transcriptional abundance of mutated loci, cohort-level baseline expression was quantified as the median RSEM Transcripts Per Million (*GeneLevelTPM*) across 371 primary solid tumor samples. Among 12,529 unique mutated genes, 11,275 had an available *GeneLevelTPM* value.

The expression profile of mutated genes exhibits a broad dynamic range spanning more than six orders of magnitude, displaying a characteristic bimodal distribution on the $\log_2(\text{GeneLevelTPM} + 1)$ scale (**Figure 2**). The left-most spike at zero represents transcriptionally quiescent or unexpressed genomic loci, while the main normal distribution peaks between $\log_2(\text{TPM} + 1)$ values of 8 and 12 (corresponding to ~250 to 4,000 TPM).

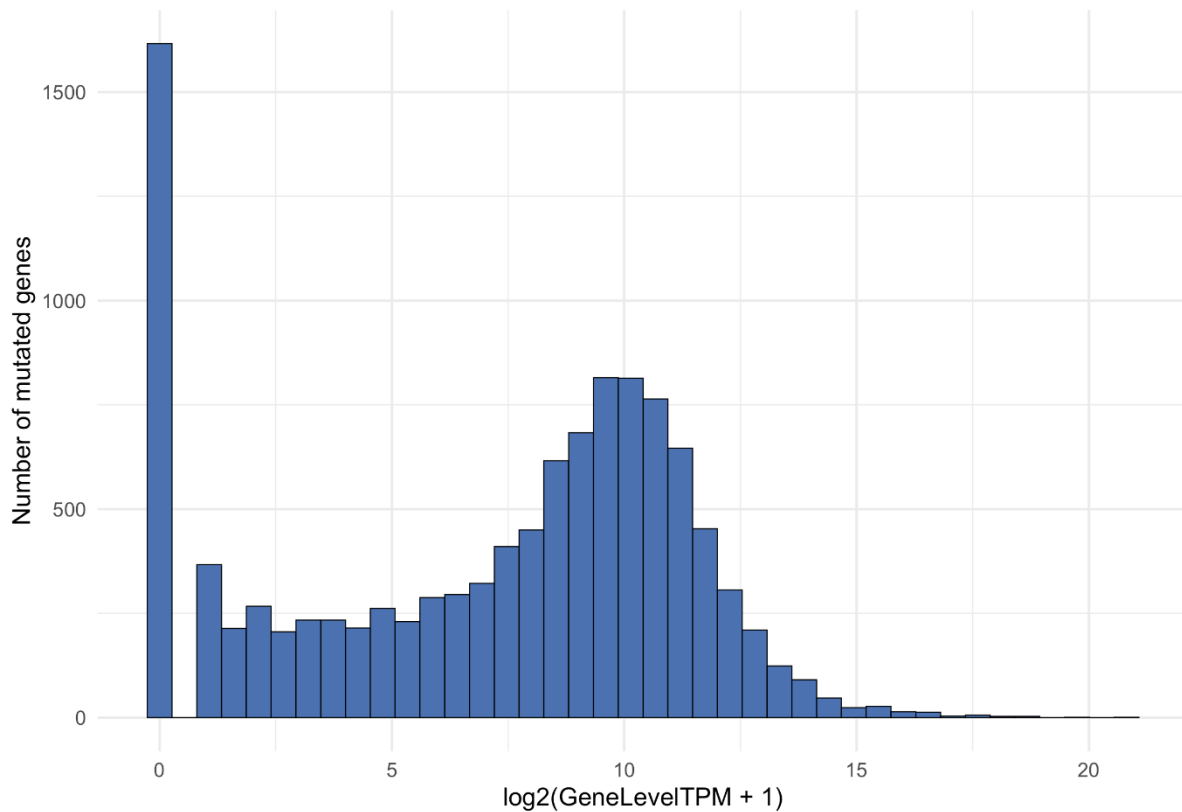


Figure 2. Distribution of median gene expression levels ($\log_2(\text{GeneLevelTPM} + 1)$) across 371 TCGA-LIHC primary solid tumor samples for all identified mutated genes ($N = 11,275$).

The top ten most highly expressed mutated genes in the cohort are heavily dominated by hepatocyte-specific serum carrier proteins, acute-phase reactants, and apolipoproteins. ALB (Albumin) exhibited the highest median expression ($\text{GeneLevelTPM} = 1,839,737.01$), followed by SERPINA1 ($\text{GeneLevelTPM} = 746,711.00$), C3 ($\text{GeneLevelTPM} = 443,757.00$), HP ($\text{GeneLevelTPM} = 421,073.00$), and FGA ($\text{GeneLevelTPM} = 379,702.00$). Additional highly abundant mutated transcripts include *APOE*, *VTN*, *TF*, *FGB*, and *APOA1* (all $\text{GeneLevelTPM} > 220,000$). This pronounced transcriptional activity confirms that key liver-specific driver loci produce sufficient mRNA abundance necessary for downstream peptide translation and antigen processing.

4.3. Neoantigen Prediction and Candidate Prioritisation

Systematic sliding-window generation across the 20 selected recurrent missense mutations yielded 1,140 peptide-HLA prediction pairs, comprising 540 9-mer Class I predictions and 600 15-mer Class II predictions. Across all tested HLA alleles, 60 mutant peptides were predicted to bind stably ($\text{IC}_{50} \leq 500$ nM or Percentile Rank $\leq 10\%$).

To isolate genuine therapeutic candidates from non-specific self-peptides, a multi-parameter filtering hierarchy identified 53 high-priority neoantigen candidates:

Table 1. Representative Top-Ranked Candidate Neoantigens in TCGA-LIHC.

Gene	Mutation	Amino Acid Change	HLA Allele	MHC Class	Mutant Peptide	Mutant IC_{50} (nM)	WT IC_{50} (nM)	Fold Change	Immunogenicity Score	GeneLevelTPM	Priority Category
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CTNNB1	c.1004A>T	p.K335I	HLA-A*02:01	I	RTYTYEILL	434.93	6,756.79	15.54	+0.234	8,899.99	Mutant-Specific Improved
	c.97T>C	p.S33P			IMRTYTYEI	102.47	12,743.07	124.36	+0.144		
					YLDPGIHSG	413.63	589.31	1.42	+0.085		
TP53	c.469G>T	p.V157F	HLA-DRB1*04:01	II	GTRFRAMAIYKQSQH	35.50	237.35	6.69	NA	1,217.00	
EEF1A1	c.1295C>T	p.T432I			MRQIVAVGVKAVDK	59.82	646.50	10.81		89,671.00	

- **Priority Tier 1 (Mutant-Specific Improved Binders, $N = 28$):** The mutant peptide binds stably while the wild-type counterpart is classified as a non-binder, minimizing central tolerance and off-target autoimmune toxicity risks.
- **Priority Tier 2 (Improved Mutant Binders, $N = 25$):** Both mutant and wild-type peptides bind, but the somatic mutation significantly enhances binding affinity ($AffinityFoldChange > 1$).

High-priority candidates were predominantly concentrated in recurrent hotspot drivers, including *CTNNB1* (p.K335I and p.S33P), *TP53* (p.V157F), and *EEF1A1* (p.T432I), coupling strong binding improvements with high tumor transcript abundance.

5. Discussion

Our findings demonstrate that high mutational recurrence alone does not guarantee effective neoantigen presentation. Across the cohort, immunogenic potential is governed by a complex interplay between mutation site, presentation pathway, and transcript abundance.

CTNNB1 was mutated in 25.6% of the analysed HCC samples. Selected candidates included Exon 3 regulatory mutations, such as p.D32V and p.S33P, as well as the recurrent p.K335I variant. Alterations at D32 and S33 may disrupt β -catenin phosphorylation and degradation, contributing to Wnt pathway activation.

Crucially, the *CTNNB1* p.K335I mutation generated two HLA-A*02:01-restricted peptides with substantially improved predicted binding relative to their wild-type counterparts. These derived epitopes exhibit marked fold-change improvements over the wild-type sequence (>15-fold to >124-fold) alongside positive TCR immunogenicity scores (+0.234). Given the robust expression of *CTNNB1* ($GeneLevelTPM \approx 8,900$), these hotspot-derived neoantigens emerge as prime candidates for personalized therapeutic cancer vaccines.

In contrast, *TP53* mutations, the single most frequent genetic lesion in the cohort (26.7%), display a distinct presentation profile. Core DNA-binding domain alterations, including p.V157F and p.R249S, showed only moderate MHC Class I affinity but generated strong Class II binding against HLA-DRB1*04:01 ($IC_{50} = 35.5$ nM). This finding underscores the necessity of profiling 15-mer Class II epitopes alongside canonical Class I peptides, as CD4+ T helper cell priming remains essential for sustaining CD8+ cytotoxic T-cell memory and counteracting the immunosuppressive tumor microenvironment in HCC. Furthermore, absolute presentation probability is strongly amplified by base gene expression, as observed in high-abundance genes such as *EEF1A1* (p.T432I, $GeneLevelTPM \approx 89,671$), where High $GeneLevelTPM$ increased the prioritisation of these candidates but does not demonstrate mutant-allele expression or cell-surface presentation.

6. Limitations

Several technical and methodological constraints should be considered when interpreting these findings. First, epitope prediction relied on a fixed 5-allele HLA panel optimized for cohort-level discovery rather than individualized patient genotyping. Translating these targets into clinical workflows will require high-resolution, four-digit patient-specific HLA typing derived directly from whole-exome sequencing. Second, while NetMHCpan provides robust predictions for peptide-MHC binding affinities, it does not explicitly model upstream proteasomal cleavage kinetics or TAP transporter efficiency, both of which can restrict the intracellular processing pipeline. Finally, these predictions remain strictly *in silico*; while binding affinities and TCR recognition metrics effectively prioritize viable candidates, spontaneous *in vivo* immunogenicity requires experimental validation through functional assays such as IFN- γ ELISpot, peptide-MHC tetramer staining, and direct identification via mass spectrometry-based immunopeptidomes.

GeneLevelTPM represents the median expression across the cohort of 371 primary tumours rather than patient-specific expression matched to each mutation. Therefore, it does not confirm that a particular mutation or mutant allele was expressed in the individual tumour carrying that mutation. The advanced analysis was restricted to 20 selected recurrent missense mutations and a predefined HLA panel, so it does not represent a comprehensive or patient-specific neoantigen analysis. All neoantigen candidates are computational predictions and require experimental validation.

7. Member Contributions

- **Anastasia Podkletnova**

Preprocessed and filtered the somatic mutation data, performed quality-control assessments, and constructed the mutation-by-sample matrix. Reviewed and refined the RNA-seq processing workflow by retrieving the RSEM TPM assay and retaining primary tumour samples. Developed the mutation–expression integration scripts, implemented the VEP-based protein annotation validation pipeline, and automated the IEDB API workflows for HLA-binding and immunogenicity prediction.

- **Hasmik Chilingaryan**

Developed the initial RNA-seq data acquisition and expression-processing workflow using curatedTCGADData. Prepared and structured the comprehensive written scientific report, documented the computational workflow and project outputs in the GitHub README, and developed the final presentation slide deck. Synthesised the project methodology, results, biological interpretation, and conclusions for the final deliverables.